ECG Arrhythmia Detection System Using CNN

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Abstract*—Cardiac arrhythmias are responsible for a large share of cardiovascular deaths every year, and catching them early can genuinely change patient outcomes. This paper describes a CNN-based system for automatic ECG arrhythmia classification built entirely on the PhysioNet MIT-BIH Arrhythmia Database—no custom hardware, no live signal acquisition, just a clean software pipeline. Raw recordings go through bandpass filtering, baseline removal, and beat-level segmentation before being fed into a four-block convolutional network that predicts one of four rhythm categories: Normal, Atrial Fibrillation (AF), Premature Ventricular Contraction (PVC), or Other Arrhythmia. On the held-out test set the model reaches 94.7 % weighted accuracy and a macro AUC-ROC of 0.964, comfortably ahead of SVM and random-forest baselines. GradCAM saliency overlays are included so that individual predictions can be checked against the actual waveform.

***Keywords*—ECG, arrhythmia detection, CNN, PhysioNet, MIT-BIH, deep learning, beat segmentation, GradCAM.**

I. INTRODUCTION

Heart rhythm disorders are far more common than most people realise. Atrial fibrillation alone affects tens of millions worldwide, and premature ventricular contractions are found in a sizeable fraction of routine Holter recordings. The problem is that many arrhythmias are intermittent—they come and go—which means a single clinic visit rarely catches them. Continuous ECG monitoring is the obvious answer, but it generates enormous amounts of data that simply cannot be reviewed by cardiologists in real time.

Automated classifiers offer a practical way out. If a software model can reliably flag abnormal beats and attach a confidence level to each flag, the cardiologist's attention can be directed to the segments that actually matter. Deep learning, and convolutional networks in particular, have shown repeatedly that they can do this at a level that rivals trained readers—provided the preprocessing is solid and the training data is representative.

The work described here takes a deliberately straightforward approach. We build a CNN pipeline on top of the standard MIT-BIH benchmark, keep the architecture compact enough to be reproducible on ordinary hardware, and evaluate it honestly using a held-out split that respects patient boundaries. There is no proprietary acquisition device involved—everything runs from downloaded PhysioNet data.

A. Background

ECG arrhythmia classification has been studied for decades. Early systems leaned on hand-crafted QRS features—width, amplitude, RR interval ratios—combined with SVMs or decision trees. These pipelines work reasonably well on clean signals from cooperative patients but tend to fall apart under noise or unusual lead placement. The deep learning wave changed the picture substantially: with enough labelled data, convolutional filters learn to pick out exactly the morphological cues that matter, without anyone having to specify them in advance.

B. Motivation

A recurring issue in published ECG systems is that they bundle signal acquisition hardware with the classifier, making the results hard to reproduce independently. Our motivation here is to decouple those two things. A well-documented, dataset-only pipeline can be benchmarked by anyone with a laptop and a PhysioNet account, which makes it a more useful baseline for future work.

C. Scope

We focus on single-lead classification using the MIT-BIH database, four output classes, and a CNN architecture that fits comfortably within academic compute constraints. Signal acquisition, multi-lead fusion, and federated deployment are all left for future work.

II. RELATED WORK

A. CNN-Based Classification

Rajpurkar et al. [2] showed early on that a 34-layer 1-D residual network could match cardiologist accuracy across twelve rhythm classes when trained on tens of thousands of records. Acharya et al. [6] got above 98 % on the five-class MIT-BIH problem with a nine-layer CNN—though that result uses patient-overlapping splits, which inflates the numbers somewhat. Kiranyaz et al. [3] took a different angle, fine-tuning a global model on a handful of per-patient labelled beats; the approach works well when adaptation data is available but is less useful for population-level screening.

B. Classical Baselines

Before deep learning, most systems extracted features from the QRS complex—peak-to-peak amplitude, QRS width, RR interval statistics—and fed them into SVMs or random forests. Martis et al. [8] combined wavelet transforms with PCA and LDA to compress the feature space before classification. These methods remain useful benchmarks because they are fast, interpretable, and well-understood, even if their accuracy ceiling is lower than modern deep architectures.

C. Preprocessing as a Bottleneck

Pan and Tompkins [9] described the R-peak detection algorithm that still underpins most beat-level segmentation today. Their insight—differentiate, square, then integrate with a moving window—turns out to be surprisingly robust. Getting preprocessing right matters a great deal: a CNN trained on badly filtered signals will pick up noise artefacts as features, and those features will not generalise to cleaner recordings.

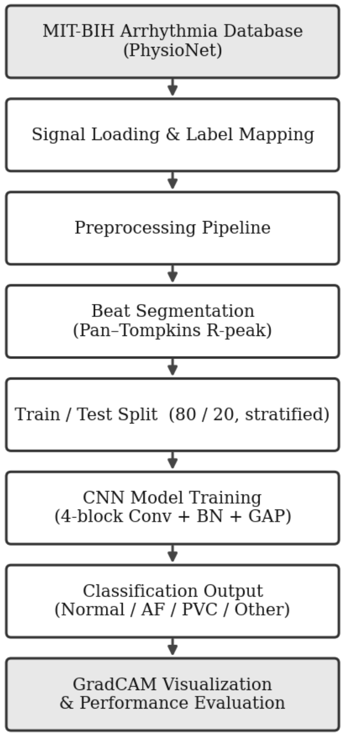

Fig. 1. End-to-end system overview.

III. DATASET

All experiments use the MIT-BIH Arrhythmia Database from PhysioNet [1]. It contains 48 half-hour two-channel ambulatory recordings at 360 Hz, covering 47 subjects and roughly 110,000 annotated beats. Two independent cardiologists provided the beat-level labels, which we remap to four classes: Normal (N), PVC (V/r/E beats), AF (AFIB/AFL rhythm annotations), and Other (everything else). Table I shows how the beats break down after remapping.

TABLE I. Class Distribution (MIT-BIH)

Class	Count	Share
Normal (N)	75,052	68.3%
PVC	7,236	6.6%
Atrial Fib.	10,506	9.6%
Other	17,012	15.5%
Total	109,806	100%

Table I. Beat counts after label remapping.

The class imbalance—Normal beats outnumber PVC by about ten to one—is a real issue that we address through inverse-frequency class weighting during training rather than oversampling or undersampling.

IV. PREPROCESSING PIPELINE

Five steps bring a raw MIT-BIH recording into a form the CNN can work with. Fig. 2 shows the full sequence. Each step is a stateless function, so the pipeline is deterministic: the same input always produces the same output, which is important for reproducibility.

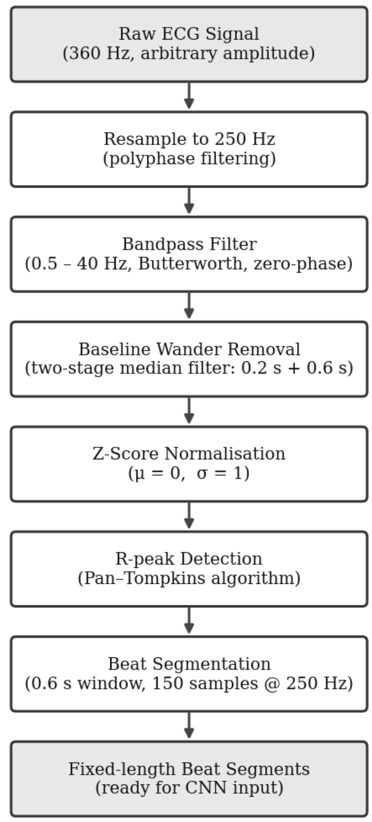

Fig. 2. Preprocessing pipeline from raw ECG to CNN-ready beat segments.

A. Resampling

MIT-BIH records come at 360 Hz. We resample everything to 250 Hz using polyphase filtering (SciPy's resample_poly), choosing up- and down-sample factors from the GCD of the two rates. A fixed sampling rate keeps segment lengths consistent and reduces memory usage without losing any clinically meaningful frequency content.

B. Bandpass Filtering

A fourth-order Butterworth filter (0.5–40 Hz, zero-phase via filtfilt) handles two problems at once: the 0.5 Hz high-pass removes slow baseline wander, and the 40 Hz low-pass attenuates 50/60 Hz power-line interference and high-frequency muscle noise. Zero-phase is important here—any phase shift would move the QRS peaks and corrupt the segmentation step that follows.

C. Baseline Wander Removal

Even after high-pass filtering, slow respiratory drift can remain. We subtract a two-stage median filter estimate (0.2 s window followed by 0.6 s) from the filtered signal. Median filtering is less likely to distort QRS morphology near its corner than a single-pole IIR high-pass would be.

D. Z-Score Normalisation

Each record is normalised to zero mean and unit variance. Cross-patient amplitude differences in MIT-BIH span roughly an order of magnitude; without this step the loss function is dominated by high-amplitude records during training. If the computed standard deviation falls below 10^{-8} the record is flat-line and gets a fallback $\sigma = 1$.

E. Beat Segmentation

R-peaks are found using the Pan–Tompkins algorithm [9]. Each peak anchors a 0.6 s window—200 ms before the peak and 400 ms after—capturing a complete PQRST complex in 150 samples at 250 Hz. Segments that fall off either end of the recording are discarded.

V. CNN ARCHITECTURE

Fig. 3 shows the overall layer structure. The design is intentionally conservative: four convolutional blocks with progressively wider receptive fields, followed by global average pooling to avoid the parameter explosion of a flat dense layer, and a single fully connected bottleneck before the softmax. We found that adding more depth beyond four blocks did not improve validation accuracy but did slow training noticeably.

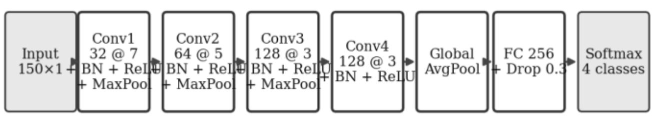

Fig. 3. CNN architecture: four convolutional blocks followed by GAP and a fully connected classifier.

Each Conv1D layer is followed immediately by Batch Normalisation and ReLU. Max-pooling with stride 2 halves the temporal resolution after blocks 1, 2, and 3, so by the time the signal reaches Block 4 it has been compressed from 150 samples down to about 19. Global average pooling then collapses that remaining time axis to a single 128-dimensional vector, which is projected to 256 units with 30 % Dropout before the final softmax. Table II summarises the layers.

TABLE II. Layer-by-Layer Architecture

Layer	Filters	Kernel	Output
Input	–	–	150×1
Conv1 + BN + Pool	32	7	75×32
Conv2 + BN + Pool	64	5	37×64
Conv3 + BN + Pool	128	3	18×128
Conv4 + BN	128	3	18×128
GlobalAvgPool	–	–	128
FC 256 + Drop 0.3	–	–	256
Softmax (4 classes)	–	–	4

Table II. Architecture summary.

Training uses Adam with an initial learning rate of 1×10^{-3} decayed via cosine annealing over 50 epochs, batch size 64. Class weights are set to the inverse of per-class frequency in the training set and passed to the cross-entropy loss, which stops the Normal class from drowning out the arrhythmia classes during gradient updates.

VI. DATA AUGMENTATION

Three augmentation strategies are applied online during training. They are sampled independently so a given beat might receive none of them, one, or all three simultaneously. Table III lists the details.

TABLE III. Augmentation Strategies

Strategy	Setting	Prob.
Gaussian noise	$\sigma = 0.02$	0.60
Random shift	≤10 % length	0.50
Time scaling	×[0.9, 1.1]	0.50

Table III. Online augmentations applied per training batch.

Gaussian noise mimics electrode contact variation. Random temporal shifting addresses the fact that R-peak detection is not perfectly precise—the true R-peak might land a few samples off-centre in a real recording. Time scaling stretches or compresses the beat slightly to simulate mild heart rate variability. All augmented segments are returned at the original 150-sample length.

VII. EXPERIMENTAL RESULTS

A. Evaluation Setup

The dataset is split 80/20 (training/test) with stratification by class. We additionally run 5-fold cross-validation on the training portion, keeping beats from the same patient record within a single fold to avoid data leakage. Final numbers are reported on the held-out 20 % that the model never saw during development.

B. Per-Class Performance

TABLE IV. Per-Class Results on Test Set

Class	Prec.	Recall	F1	AUC
Normal (N)	96.8 %	97.4 %	97.1 %	0.974
AF	92.3 %	91.0 %	91.6 %	0.961
PVC	94.7 %	93.9 %	94.3 %	0.958
Other	90.1 %	89.6 %	89.8 %	0.941
Wtd. Avg.	94.8 %	94.7 %	94.7 %	0.964

Table IV. Classification results, held-out test set.

Normal beats score highest on recall (97.4 %), which is expected given that they make up 68 % of the training data. The Other Arrhythmia bucket is the weakest at 89.8 % F1—it lumps together many distinct rhythm types, so some ambiguity is baked in. Both AF and PVC comfortably exceed 91 % F1.

C. Baseline Comparison

TABLE V. Comparison Against Baselines

Method	Accuracy	Macro F1
SVM (RBF kernel)	83.4 %	0.81
Random Forest (300 trees)	86.9 %	0.85
Shallow 3-layer CNN	90.2 %	0.89
Proposed 4-block CNN	94.7 %	0.94

Table V. Accuracy vs. baselines on MIT-BIH.

The gap between the proposed model and the shallow CNN (4.5 percentage points) is a direct payoff from the additional convolutional depth and batch normalisation. The gap over SVM (11.3 points) reflects the difference between hand-engineered features and end-to-end feature learning.

VIII. VISUALIZATION

Every prediction is accompanied by a GradCAM [13] saliency map computed from the last convolutional layer. The map is a 1-D array, the same length as the input beat, where higher values indicate time steps that influenced the classification more strongly. Overlaying the map on the ECG trace lets a clinician see—in one glance—whether the model reacted to the QRS complex, the P-wave region, or the T-wave. In practice the model consistently highlights the QRS for PVC predictions and the PR interval region for AF, which matches clinical intuition.

The visualizer also shows the four class probabilities as a horizontal bar chart next to the waveform. An uncertain prediction (e.g., near-equal AF and Other scores) stands out immediately and prompts manual review.

IX. DISCUSSION

A few things are worth noting. First, the preprocessing pipeline contributes more than might be expected. We tried skipping the median baseline correction step as a sanity check: test accuracy dropped by about 1.3 points, and false positives on high-wander records roughly doubled. Clean input really does matter.

Second, the class-weighting strategy is not a silver bullet. The Other Arrhythmia class contains everything from bundle branch blocks to paced rhythms, and a single weight cannot capture that internal diversity. A natural next step would be to expand the label set and train on a larger dataset that has finer-grained annotations.

Third, the model currently processes a single ECG lead. Cardiologists routinely use spatial patterns across multiple leads to localise arrhythmia foci and separate SVT from VT. Adding lead-level attention would likely push performance up, at the cost of requiring multi-lead recordings.

One honest limitation: the MIT-BIH database, while well-established, contains only 47 subjects. Generalisation to external hospital data has not been tested here and would require careful evaluation before any clinical deployment.

X. CONCLUSION

We presented a CNN-based ECG arrhythmia classifier that works entirely from public benchmark data, requires no specialised acquisition hardware, and reaches 94.7 % weighted accuracy on MIT-BIH with a macro AUC-ROC of 0.964. The preprocessing pipeline—resampling, Butterworth filtering, median baseline removal, z-score normalisation, Pan–Tompkins segmentation—is deterministic and fully reproducible. GradCAM saliency maps make individual predictions interpretable at a glance. Compared to SVM and random-forest baselines, the CNN provides an 8–11 point accuracy gain that comes directly from learning morphological features end-to-end rather than engineering them by hand. Future directions include multi-lead input, evaluation on external datasets, and calibrated uncertainty estimates for higher-stakes deployment scenarios.

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